

1. Executive Summary & Solution Architecture

The **Maintenance Request System (MRS)** is an enterprise-grade hybrid web application developed with **ASP.NET Core 10**. It combines a rich server-side **MVC Razor Web Portal** with a high-performance, secure **RESTful Web API (v1)** to automate and oversee facility repair workflows across corporate departments.

100%
SHARED LOGIC

.NET 10
FRAMEWORK TARGET

20+ Endpoints
REST WEB API V1

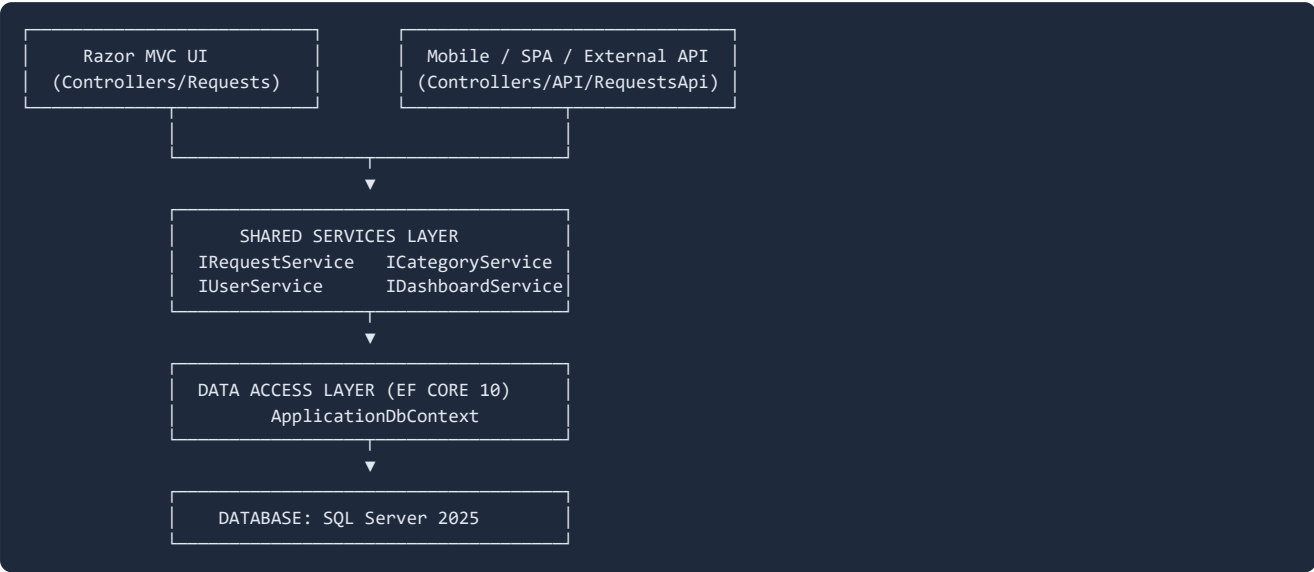
Enterprise Architecture Matrix

Dimension	ASP.NET Core MVC Layer	RESTful Web API Layer (/api/v1)
Target Clients	Desktop/Laptop Web Browsers (Chrome, Edge, Safari)	Mobile Apps (Flutter, React Native), SPAs (React, Angular), External Systems
UI Presentation	Bootstrap 5 Razor Views with server-rendered HTML	Headless JSON contracts using standardized ApiResponse<T> envelopes
Authentication	ASP.NET Core Identity Application Cookie with AntiForgery	Stateless JWT Bearer Tokens with customizable claims
Business Logic	Delegates completely to IRequestService, ICategoryService	Delegates to the exact same Services (Zero code duplication)

- Key System Capabilities
- Unified Domain Services:** Single source of truth for status transitions, ticket lifecycle, validation, and audit recording.
 - Role-Based Access Control:** Strict segregation of duties for Administrator, Field Technician, and Corporate Employee.
 - Automated Work Order Numbering:** Generates formatted identifiers (REQ-YYYY-NNNN) uniquely tracked in SQL Server.
 - Full Audit Trail History:** Chronological event logging for creations, assignments, status changes, and repair notes.

2. Clean Architecture & Code Organization

The codebase is designed strictly adhering to Clean Architecture and Separation of Concerns principles. Both presentation mechanisms (MVC and API) consume shared services, eliminating divergent business rules.



Solution Structure & Component Roles		
Directory	Key Classes	Architectural Responsibility
Controllers/	Account, Requests, Dashboard, Categories, Users	Handles browser navigation, view binding, model states, TempData notifications.
Controllers/API/	AuthApi, RequestsApi, CategoriesApi, UsersApi	Produces standardized JSON outputs, handles HTTP status codes and API versioning.
Services/	RequestService, CategoryService, UserService	Encapsulates validation, state machine rules, query filtering, and audit history.
DTOs/	CreateRequestDto, RequestDetailsDto, AuthResponseDto	Decouples internal EF Core entities from client payloads, preventing over-posting.
Data/	ApplicationDbContext, Migrations	Configures fluent relationships, soft-delete query indexes, and seed datasets.
Middleware/	GlobalExceptionMiddleware	Intercepts unhandled errors: outputs JSON for API callers and error view for MVC.

3. Security, Identity & Dual Authentication

MRS employs a unified security model that concurrently satisfies web browser sessions and distributed API consumers via hybrid authorization policies registered in `Program.cs`.



ASP.NET Core Identity (MVC)

Uses secure HTTP-only cookies configured with sliding expirations, Anti-Forgery Tokens (CSRF protection), and lockout protection on repetitive invalid login attempts.

- Cookie Path: `/Account/Login`
- Lockout: 5 failed attempts locks for 5 min
- Roles: Admin, Technician, Employee



JWT Bearer Authentication (API)

Issues HMAC-SHA256 cryptographically signed tokens containing user identity, claims, and assigned role credentials for stateless mobile/client consumption.

- Issuer: `MaintenanceRequestSystem`
- Expiration: 60 minutes configurable
- Header: `Authorization: Bearer <Token>`



Multi-Tier Security Controls Matrix

Security Mechanism	Implementation Detail	Threat Mitigated
Rate Limiting	Fixed-window limiter: 5 attempts/min on <code>/api/v1/auth/*</code>	Brute force & Credential stuffing attacks
Resource Authorization	Service checks <code>request.EmployeeId == currentUserId</code>	Insecure Direct Object References (IDOR)
Data Sanitization	DTOs omit password hashes, concurrency stamps, and internal flags	Sensitive data exposure
CORS Policy	Configurable allowed origins (Angular, React, Vue local ports)	Unauthorized Cross-Origin requests

```
// Dual Authentication Policy in ServiceExtensions.cs
services.AddAuthorization(options => {
    options.DefaultPolicy = new AuthorizationPolicyBuilder(
        IdentityConstants.ApplicationScheme,
        JwtBearerDefaults.AuthenticationScheme)
        .RequireAuthenticatedUser()
        .Build();
});
```

4. Database Schema & Domain Modeling

The database layer is managed via **EF Core 10 Code-First** on **SQL Server 2025**. It enforces referential integrity, domain constraints, soft deletion, and optimal query indexing.

Relational Domain Schema Definition

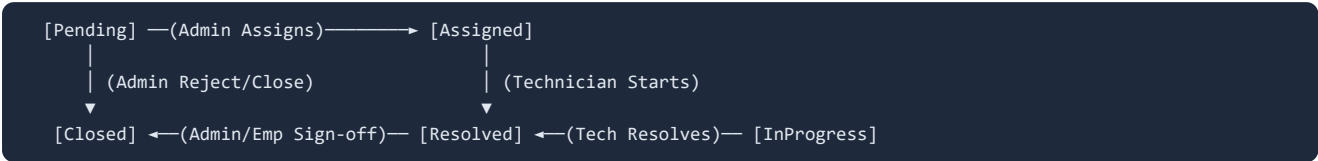
Entity Table	Key Columns & Types	Relationships & Cascade Rules
ApplicationUser	Id (string PK), FullName, Department, IsActive, LastLoginAt	1-to-Many with MaintenanceRequest, Comments, and Audit Histories.
MaintenanceRequest	Id (int PK), RequestNumber (string UK), Title, Description, Status, Priority, CategoryId (FK), EmployeeId (FK), TechnicianId (FK)	Foreign keys configured with DeleteBehavior.Restrict to prevent accidental cascaded deletion of critical audit history.
Category	Id (int PK), Name, Description, IsActive	Pre-seeded with Electrical, Plumbing, HVAC, IT Equipment, Furniture.
RequestComment	Id (int PK), CommentText, CreatedAt, RequestId (FK), UserId (FK)	Enables ongoing technical dialogue between employee and technician.
RequestHistory	Id (int PK), Action, Description, CreatedAt, RequestId (FK), UserId (FK)	Immutable append-only audit trail logging each status mutation and assignment.

Indexing & Soft Delete Architecture

- Unique Business Key:** IX_MaintenanceRequests_RequestNumber guarantees uniqueness for ticket lookups.
- Soft Delete Query Optimization:** Composite index on (IsDeleted, Status) enables efficient server-side filtering of active records.
- Soft Delete Strategy:** When a request is deleted, IsDeleted = true and DeletedAt = UtcNow are set, preserving historical logs without data loss.

5. Business Logic & Ticket Lifecycle State Machine

Ticket states strictly advance according to an automated finite state machine. Invalid transitions are rejected at the service layer, guaranteeing process compliance across both UI and REST API.



State Transition Rules Matrix

From Status	To Status	Authorized Roles	Validation & Business Rules
Pending	Assigned	Admin	Assigns active Technician ID; logs TechnicianAssigned history.
Pending	Closed	Admin	Immediate dismissal of invalid or duplicate work orders.
Assigned	InProgress	Technician (Assigned)	Technician marks commencement of on-site inspection/work.
InProgress	Resolved	Technician (Assigned)	Requires mandatory ResolutionNotes; stamps ResolvedAt.
Resolved	Closed	Admin / Employee (Author)	Final verification and sign-off; stamps ClosedAt timestamp.

Role Permission Boundaries

- Employee:** Can only update or delete requests in Pending status. Cannot reassign technicians or alter statuses.
- Technician:** Can only modify status of requests explicitly assigned to them. Cannot access admin user management.
- Admin:** Global oversight to assign technicians, configure equipment categories, and manage staff user accounts.

6. RESTful Web API Specifications (/api/v1)

The REST API delivers standardized endpoints with explicit HTTP status codes, query filtering, pagination caps, and predictable envelope responses.

REST Endpoints Catalog

Method	Endpoint	Auth / Role	Description & Behavior
POST	/api/v1/auth/login	Public	Authenticates credentials, returns JWT token and user roles.
POST	/api/v1/auth/register	Public	Registers new Employee/Technician account.
GET	/api/v1/requests	Authenticated	Paginated search (?page=1&pageSize=10&priority=High&status=Pending).
POST	/api/v1/requests	Employee	Creates new ticket. EmployeeId extracted securely from JWT claims.
GET	/api/v1/requests/{id}	Resource Allowed	Returns full details, category, employee, technician, comments, history.
POST	/api/v1/requests/{id}/assign	Admin	Assigns a technician to the request and sets status to Assigned.
PUT	/api/v1/requests/{id}/status	Technician / Admin	Transitions request status and attaches resolution notes.
GET	/api/v1/requests/{id}/comments	Resource Allowed	Fetches chronological technical comments for a work order.
GET	/api/v1/dashboard/admin	Admin	Aggregated metrics: requests by status, priority, category.

Standardized Response Contract (ApiResponse<T>)

```
{
  "success": true,
  "message": "Request retrieved successfully",
  "data": { "id": 2, "requestNumber": "REQ-2026-0002", "status": "Resolved" },
  "errors": null
}
```

7. User Interface: Authentication & Admin Dashboard

Live captures from the running application illustrating the modern, accessible design built with Bootstrap 5 and customized Razor templates.

 Screen 7.1: Portal Login Screen (/Account/Login)

Clean authentication portal with pre-configured quick-fill demo credentials and input validation feedback.



Maintenance Request System

Sign In

Email Address



 Screen 7.2: Administrator Central Dashboard (/Dashboard)

Aggregated operational KPIs showing total volume, pending dispatches, resolved tickets, and active staff counts.


Maintenance Request System

[admin@maintenance.com](#)
[Admin](#)
[API Docs](#)
[Logout](#)

Dashboard

Requests

Categories

Users

Admin Dashboard

Saturday, September 5, 2026

3
Total Requests

2
Pending

0
In Progress

1
Resolved

0
Closed

0
Critical

4
Total Users

1
Technicians

Recent Requests

View All

Request #	Title	Status	Priority	Employee	Created
1001	Replace broken light bulb in hallway	Pending	Low	John Doe	2023-09-05 10:30
1002	Fix leaking faucet in kitchen	In Progress	Medium	Jane Smith	2023-09-05 11:15
1003	Repair damaged door handle	Resolved	Low	John Doe	2023-09-05 12:45
1004	Check and replace air filter	Pending	Medium	Jane Smith	2023-09-05 13:20
1005	Inspect and tighten loose screws	Closed	Low	John Doe	2023-09-05 14:00

8. Work Order Management & Audit Trail

Live captures demonstrating the comprehensive ticket management capabilities, real-time filtering, and chronological event logging.

Screen 8.1: Requests Catalog & Filter Grid (/Requests)

Multi-parameter filtering by Status, Priority, Category, and text keyword search with server-side pagination.

Maintenance Request System

admin@maintenance.comAdminAPI DocsLogout

Dashboard

Requests

Categories

Users

Maintenance Requests

Search...

All Status

All Priority

All Categories

Q

X

Showing 1 - 3 of 3 requests

Request #	Title	Status	Priority	Category	Employee	Technician	Created
REQ-2026-0003	AC not cooling	Pending	Medium	HVAC	Sara Mohamed	-	Sep 5, 2026

Screen 8.2: Detailed Ticket View, Comments & Timeline (/Requests/Details/2)

Comprehensive ticket inspection displaying status controls, technician notes, comments feed, and audit history.

Maintenance Request System

admin@maintenance.comAdminAPI DocsLogout

Dashboard

Requests

Categories

Users

REQ-2026-0002

Created September 5, 2026 at 07:22

DeleteBack

Request Details

Printer paper jam

Office printer has constant paper jam issues

STATUS

Resolved

CATEGORY

IT Equipment

TECHNICIAN

PRIORITY

High

EMPLOYEE

Sara Mohamed

RESOLVED AT

History

RequestCreated

Maintenance request created
Sara Mohamed — Sep 5, 07:22

TechnicianAssigned

Technician assigned to request
System Administrator — Sep 5, 07:22

StatusChanged

Status changed to InProgress
Ahmed Hassan — Sep 5, 07:22

Maintenance Request System Architecture

CONFIDENTIAL & PROPRIETARY

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9. Role Workflows & Administrative Management

Captures showing employee request creation forms alongside administrative consoles for categories and system users.

Screen 9.1: New Request Form (/Requests/Create)

Simple ticket creation interface for employees with validation.

Maintenance Request System

emp@maintenance.com Employee API Docs Logout

Dashboard Requests New Request

New Maintenance Request

Title *
Brief description of the issue

Description *
Detailed description of the issue...

Category *
Select category...

Priority *
Low

Submit Request Cancel

Screen 9.2: Category Management (/Categories)

Admin configuration for maintenance equipment types.

Maintenance Request System

admin@maintenance.com Admin API Docs Logout

Dashboard Requests Categories

Categories

New Category

Name	Description	Requests	Active	Created	
Electrical	Electrical systems and wiring issues	0	Active	Jan 1, 2025	
Furniture	Furniture repair and replacement	0	Active	Jan 1, 2025	
HVAC	Heating, ventilation, and air conditioning	1	Active	Jan 1, 2025	
IT Equipment	Computers, printers, and network equipment	2	Active	Jan 1, 2025	
Plumbing	Water and drainage issues	0	Active	Jan 1, 2025	

Screen 9.3: User Directory & Account Controls (/Users)

Admin console to oversee staff members, roles, activity statuses, and activate/deactivate accounts.

Maintenance Request System

admin@maintenance.com Admin API Docs Logout

Dashboard Requests Categories Users

User Management

4 Users

Name	Email	Department	Roles	Requests	Status	Actions
Ahmed Hassan	tech@maintenance.com	Maintenance	Technician	0	Active	
Ahmed Tester	emp2@maintenance.com	-	Employee	0	Active	
Sara Mohamed	emp@maintenance.com	HR	Employee	3	Active	
System Administrator	admin@maintenance.com	IT	Admin	0	Active	

